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# Lab 5

## Fluid Dynamics

## Abstract

This lab looked at several concepts of fluid mechanics using a closed-loop wind tunnel. The experiments focused on wind tunnel calibration, vortex shedding from a circular cylinder, wake momentum loss and drag, boundary layer development, airfoil performance, and the differences between two and three dimensional bluff body flow. A pitot static tube was used to relate tunnel fan frequency to freestream velocity, which allowed velocity prediction at unmeasured tunnel frequencies. Vortex shedding frequencies were measured using an accelerometer and related to Reynolds and Strouhal numbers. Wake measurements were used to calculate the drag from momentum deficit, while the boundary layer rakes were used to look at velocity profiles near a wall. Lift, drag, and pitching moment were used to measure a Clark-Y airfoil across angles of attack and Reynolds numbers, including clean and dirty configurations, to look at stall behavior. Lastly, cylinders of varying length were tested to look at three dimensional end effects on drag. Overall, the results followed theory and showed how viscous effects, flow separation, and Reynolds numbers influence aerodynamic behavior.

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## Introduction & Background

### Nomenclature

|                                     |                                    |
|-------------------------------------|------------------------------------|
| $A$ = Area                          | $M$ = Pitching Moment              |
| $A_p$ = Planform Area of Airfoil    | $m$ = Meters                       |
| $\alpha$ = Angle of Attack          | $\mu$ = Dynamic Viscosity          |
| $c$ = Chord                         | $N$ = Newtons                      |
| $D$ = Diameter                      | $\hat{n}$ = Normal Position Vector |
| $F_D$ = Drag Force                  | $P$ = Pressure                     |
| $F_L$ = Lift Force                  | $\rho$ = Density                   |
| $f$ = Vortex Shedding Frequency     | $\pi$ = 3.1416 . . .               |
| $g$ = Gravitational Acceleration    | $Re$ = Reynolds Number             |
| $Hz$ = Hertz                        | $St$ = Strouhal's Number           |
| $h = \Delta z$ = Change in Height   | $s$ = Seconds                      |
| $km$ = Kilometers                   | $U$ = Velocity Magnitude           |
| $u$ = Downstream Velocity           | $w$ = Width                        |
| $u_\infty$ = Free Stream Velocity   | $\frac{L}{D}$ = Lift to Drag Ratio |
| $V$ = Volume                        | $v$ = Velocity                     |
| $y$ = Direction Along Pressure Rake |                                    |

*These terms may be redefined throughout the report as needed.*

This lab's goal was to connect fundamental fluid mechanic theory to real experimental measurements using a wind tunnel. Flows are strongly influenced by boundary layers and flow separation, which play a large role in determining aerodynamic forces such as lift and drag, namely for bluff bodies and airfoils operating near stall. Principles of conservation of mass and momentum change how flow moves through a wind tunnel and around objects, which Bernoulli's equation shows relationships between pressure and velocity. The Reynolds number serves as a way to measure the importance of forces, which allows flow behavior to be compared across different speeds and geometries. Each experiment highlights different aspects of these principles.

### Basic Information and assumptions

This lab requires the understanding of basic concepts outlined in fluid dynamics. The following assumptions are made for the experimentation done:

Through the use of the following continuity equation, we assume that mass is conserved throughout the wind tunnel:

$$\frac{\delta}{\delta t} \int_{CV} \rho dV + \int_{CS} \vec{v} \cdot \vec{n} dA$$

Similarly, we assume momentum is conserved in the wind tunnel with the following equation:

$$\frac{\delta}{\delta t} \int_{CV} \vec{v} \rho dV + \int_{CS} \vec{v} \rho \vec{v} \cdot \vec{n} dA = \Sigma \vec{F}$$

Taking the integral of the momentum of conservation and converting to differential form using the Reynolds transport theorem and stress decomposition, then removing density and substituting  $\nu = \frac{\mu}{\rho}$  leaves us with the derived Navier-Stokes Equations:

$$\frac{\delta \vec{v}}{\delta t} + (\vec{v} \cdot \nabla) \vec{v} = gh - \frac{1}{\rho} \nabla P + u = \nu \nabla^2 \vec{v}$$

Finally, we look at the Reynolds number, which is a dimensionless parameter measuring the wind tunnel's flow over its viscous forces. This is defined as:

$$Re = \frac{\rho U D}{\mu}$$

The Reynolds number separates inertial and viscous forces in a flow, which is why it is useful for interpreting experimental results. A high Reynolds number causes inertial effects to dominate, meaning higher flow velocities and therefore a greater tendency for the flow to become turbulent. At a low Reynolds number, viscous forces are more important and the flow stays smooth and therefore laminar. For experiments involving flow over a circular cylinder, such as the experimentation completed in this lab, it is generally understood that turbulence begins to appear at around  $Re \approx 3 \cdot 10^5$ .

## Experiment 1: Wind Tunnel Calibration

In order to begin experimentation, we must be able to derive an equation for the velocity of the wind tunnel based on the fan speed. We must create a calibration equation to relate the two. Figure 1.1 shows a pitot tube, a thin and hollow tube with two taps in it used to measure total and static pressure of the flow. The hole at the end of each tube brings the flow to zero velocity to measure total pressure, while the holes on the side are used to measure static flow.

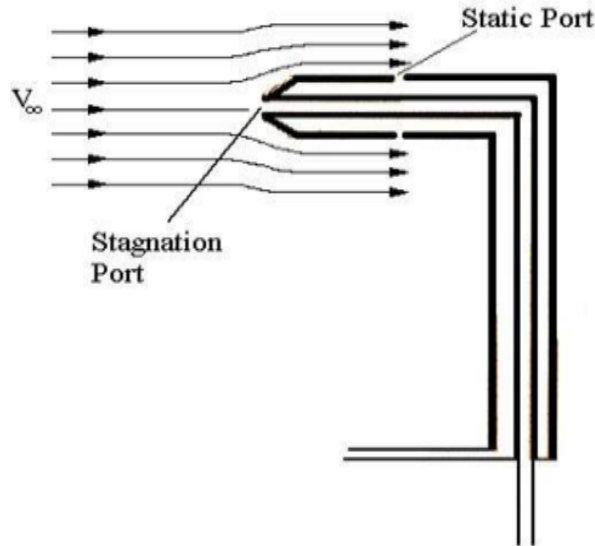


Figure 1.1: Pitot-Static Tube Diagram<sup>[1]</sup>

Bernoulli's equation can be used to relate static and dynamic pressures. These pressures can be used to determine total pressure also through Bernoulli's equation, defined as:

$$P_1 + \frac{1}{2}\rho U_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho U_2^2 + \rho gh_2$$

We can find this total pressure by simplifying this equation due to the fact that speeds are noticeably less than mach, therefore allowing us to assume that because the flow is incompressible, steady, inviscid, and irrotational, and because the wind tunnel minimizes turbulence so the flow follows streamlines, the pressure measurements can be related directly using Bernoulli's equation. Under these assumptions, we derive that:

$$P + \frac{1}{2}\rho U^2 + \rho gh = \text{const.}$$

This is further simplified because our wind tunnel assumes that gravity is neglected as we have near zero height difference, allowing us to remove the last part of the equation leaving us with:

$$P + \frac{1}{2}\rho U^2 = \text{const.}$$

Where static pressure is defined as:

$$P_{stat} = P$$

Dynamic pressure is defined as:

$$P_{dym} = \frac{1}{2}\rho U^2$$

And the total pressure, also known as stagnation pressure, can be found from our equations derivation:

$$P_{stat} = P + \frac{1}{2}\rho U^2$$

Now, we derive to find the velocity of the system by subtracting the static pressure from the total pressure and dividing both sides by  $\frac{1}{2}$  and  $\rho$ , finally taking the square root of both sides to leave us with velocity. This process is shown below:

$$P_{stag} - P_s = \frac{1}{2} \rho U^2$$

$$\frac{2(P_{stag} - P_s)}{\rho} = U^2$$

$$U = \sqrt{\frac{2(P_{stag} - P_s)}{\rho}}$$

## Experiment 2: Von Kármán Vortex Shedding

In introductory fluid analysis, flow is often assumed to be steady, inviscid, and irrotational, allowing the Navier-Stokes equations to reduce Bernoulli's equation. While these are useful assumptions for understanding the basics of flow behavior, they are not common in the real world and break down when viscous effects and flow separation are important. The Von Kármán vortex shedding experiment introduces these unsteady aerodynamics and demonstrates how simple geometries can produce time-dependent flow behavior.

When a uniform flow passes over a bluff body such as a circular cylinder, the boundary layer separates from the surface due to adverse pressure gradients. At a high Reynolds number, this separation becomes unsteady and leads to alternating shedding of vortices from opposite sides of the cylinder. The repeating pattern of vortices downstream of the body is known as a Von Kármán vortex street, and can be seen in Figure 1.2. For circular cylinders, vortex shedding occurs over a wide Reynolds number range, approximately  $47 < Re < 10^7$ , with a well defined and repeatable shedding frequency over most of this range.

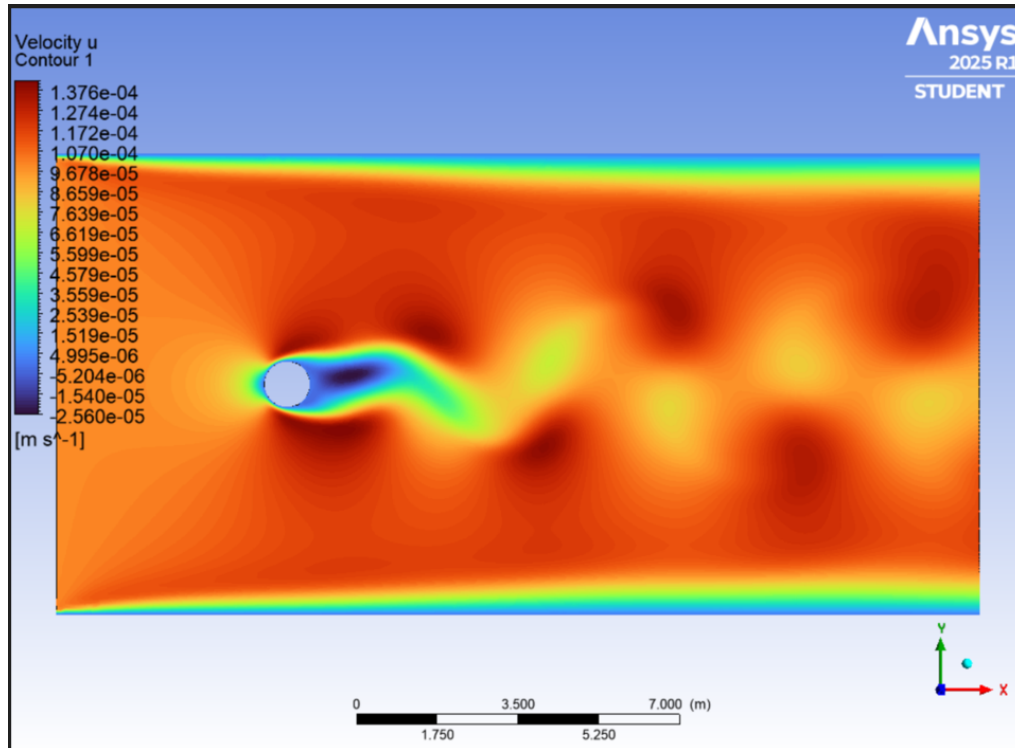


Figure 1.2 Von Karman Vortex Street shed from cylinder (top down view) in Ansys Fluent Student Edition.<sup>[2]</sup>

The vortex shedding's alternations create asymmetry in pressure distribution around the cylinder, producing oscillations perpendicular to the flow direction. These unsteady forces can form vibrations in the structure. If the vortex shedding frequency approaches the natural frequency of the cylinder, resonance can occur, creating large vibration amplitudes. This explains real world observations, such as the behavior of golf balls.

For a smooth sphere, the boundary layer near the front remains laminar and slowly loses energy as it moves down the ball due to friction. A smooth golf ball develops a laminar boundary layer at the front, which thickens as it moves downstream and quickly loses energy. Once the flow passes the halfway point, it can no longer stay attached and separates early, creating a large wake with strong unsteady vortices that increase pressure drag and slow the ball down. Adding dimples forces the boundary layer to become turbulent, allowing it to mix with the surrounding air and keep more energy. This higher energy flow stays attached farther around the back of the ball, pushing the separation point downstream and shrinking the wake, thus weakening the vortex shedding. Although the dimples slightly increase friction drag, the reduction in pressure drag is much larger, which is why a dimpled golf ball can travel about twice as far as a smooth one.<sup>[3]</sup>

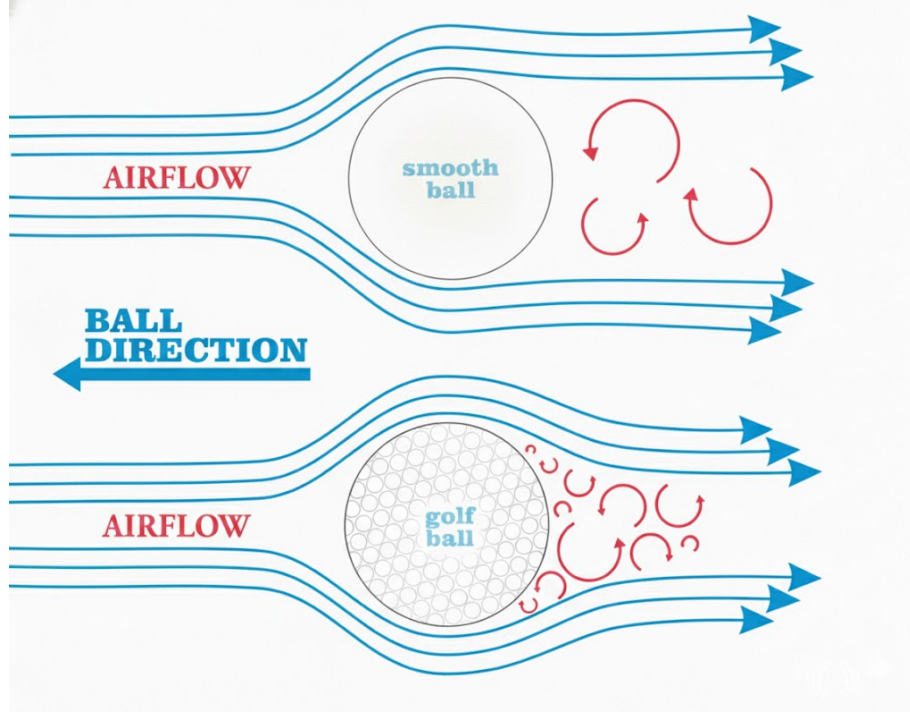


Figure 1.4: Comparison of a Smooth Ball versus Golf Ball in steady airflow <sup>[4]</sup>

The vortex shedding frequency is defined using the Strouhal number, a dimensionless parameter defined as:

$$St = \frac{fD}{U_{\infty}}$$

With  $f$  being the vortex shedding frequency,  $D$  as the cylinder diameter, and  $U_{\infty}$  being the freestream velocity. For a circular cylinder, the Strouhal number is approximately constant over a wide Reynolds number range. An empirical relationship for the shedding frequency in the range  $250 < Re < 2 \cdot 10^5$  is given by:

$$\frac{fD}{U_{\infty}} = 0.198 \left( 1 - \frac{19.7}{Re} \right)$$

In this experiment, a cylinder is mounted in the wind tunnel with an accelerometer attached to measure its vibrational response. By using the wind tunnel at different speeds and recording its acceleration data, the vibrational frequency of the cylinder can be connected to its vortex shedding frequency. This setup allows the unsteady aerodynamic behavior that's predicted by vortex shedding theory to be measured through vibration.

### Experiment 3: Wake

In this experiment, the wake behind a bluff body is examined to show how flow separation leads directly to drag and momentum loss. As the flow moves over the bluff body, the boundary layer separates due to the creation of a low pressure wake downstream while a high pressure stagnation region forms at the front of the body. This pressure difference is the main source of pressure drag. The separated flow in the wake has lower velocity than the freestream, resulting in a momentum deficiency that can be seen as a velocity loss behind the body. Downstream, the static pressure in the wake is significantly reduced, but as the flow mixes with the surrounding freestream, the pressure gradually recovers and approaches the freestream value at a distance of roughly 10 diameters downstream. The magnitude of the momentum deficiency, and therefore drag, is determined by comparing the mass and momentum flux in the flow upstream of the bluff body to that in the wake region downstream.

Real world application of analyzing wake can be seen when trying to minimize drag on vehicles and aircraft, predicting the vibration of structures due to vortex shedding, and improving the efficiency of wind farms by considering how wakes impact downstream turbines. Engineers can use experimental data gathered from understanding how flow separates and the sizes of wake to make decisions about the projects they create.

In this experiment, the wake is studied using a series of cylinders mounted perpendicular to the freestream in a wind tunnel. Drag measurements are taken using a force balance connected to the cylinders while the tunnel is operated at set speeds of 25 Hz and 45 Hz.

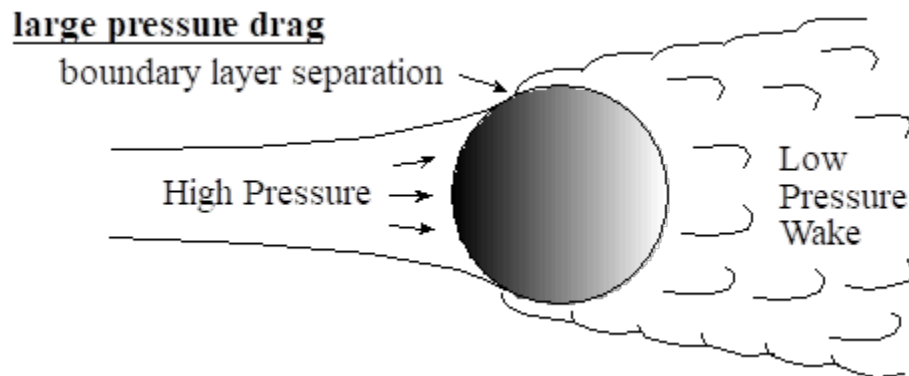


Figure 1.5: Diagram of Wake Behind a Bluff Body <sup>[5]</sup>

The formula to calculate this drag is:

$$F_D = \int_0^w \rho u_2 (u_1 - u_2) dy$$

Where  $\rho$  is the density of the air,  $u_1$  is the freestream velocity,  $u_2$  is the velocity behind the cylinder,  $w$  is the width of the control volume, and  $y$  is the direction along the pressure rake. In order to derive this equation, we must start with a model for the control volume for flow over a cylinder. The momentum equation and conservation of mass for an inertial control volume must then be combined. This process is described below.

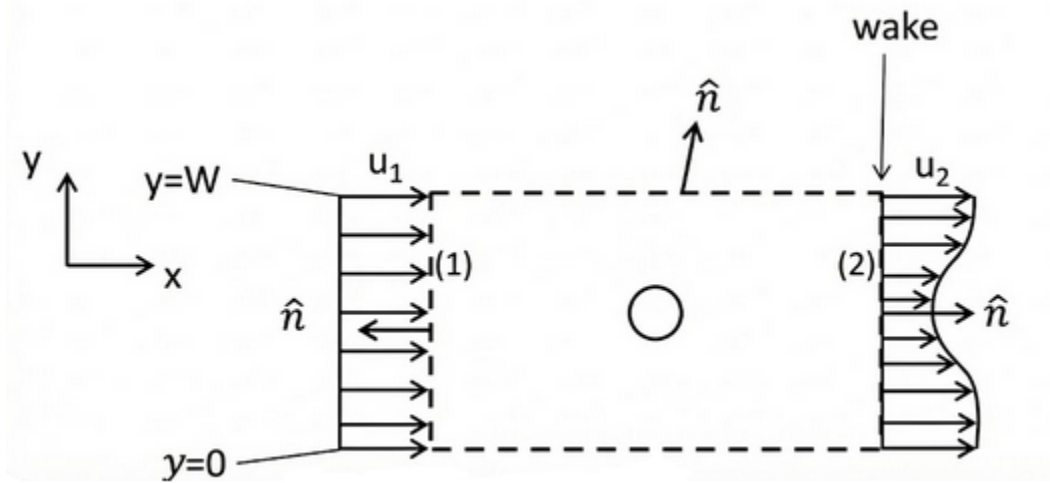


Figure 1.6: Birds-Eye View of a Control Volume for Flow over a Cylinder <sup>[6]</sup>

An inertial control volume is drawn around the circular cylinder, going upstream and downstream far enough for the pressure on the inlet and outlet surfaces to be equal to the pressure. The flow is assumed steady and incompressible, and the volume has depth into the page, so the differential area reduces to  $dA = dy$ .

Momentum equation for an inertial control volume:

$$\sum \vec{F} = \frac{\partial}{\partial t} \int_{CV} \vec{v} \rho dV + \int_{CS} \rho \vec{v} \cdot \hat{n} dA = - \int_{CS} P dA - F_D$$

Because the flow is steady and therefore unaffected by time, the time derivative is zero. Pressure forces on the inlet and outlet cancel since the static pressure is the same on both surfaces. The only remaining external force in the x-direction is the drag force exerted by the cylinder on the fluid, thus:

$$\sum \vec{F}_x = \int_{CS} \rho \vec{v} \cdot \hat{n} dA = - F_D$$

Surface 1: The outward normal points opposite the flow direction, so:

$$\vec{v} \cdot \hat{n} = - u_1$$

Surface 2: The outward normal points in the direction of the flow so:

$$\vec{v} \cdot \hat{n} = u_2$$

Substitute these two equations into the momentum equation in x-direction:

$$-F_D = \int_0^w u_1 \rho (-u_1) dy + \int_0^w u_2 \rho u_2 dy$$

Rearrange:

$$F_D = \int_0^w \rho u_1^2 dy - \int_0^w \rho u_2^2 dy$$

*This expression shows that drag is equal to the difference between the incoming and outgoing stream.*

Conservation of mass for an inertial control volume:

$$\frac{d}{dt} \int_{CV} \rho dV + \int_{CS} \rho \vec{v} \cdot \hat{n} dA = 0$$

Because the flow is steady and therefore unaffected by time, the time derivative is zero:

$$\int_{CS} \rho \vec{v} \cdot \hat{n} dA = 0$$

Looking at this over the inlet and outlet surfaces gives this as the conservation of mass equation:

$$-\int_0^w \rho u_1 dy + \int_0^w \rho u_2 dy = 0$$

Multiplying the mass conservation equation by  $u_1$  gives:

$$\int_0^w -\rho u_1^2 dy + \int_0^w \rho u_1 u_2 dy = 0$$

Substituting the mass conservation result into the drag expression gives:

$$F_D = \int_0^w \rho u_1 u_2 dy - \int_0^w \rho u_2^2 dy$$

Factoring the integrand gives the final drag form:

$$F_D = \int_0^w \rho u_2 (u_1 - u_2) dy$$

This final equation shows that drag comes from the momentum loss in the wake as wherever the downstream velocity,  $u_2$ , is less than the upstream velocity,  $u_1$ , momentum is lost.

## Force Balance, Calibration, and Data Reduction <sup>[1]</sup>

A force balance measures aerodynamic loads by detecting strain within the load cell and converting it into voltage signals via strain gauges. Recurring calibration is necessary to correct sensor drift; which is done by applying known weights to specific axes to generate linear calibration curves. For this setup, a six-degree-of-freedom, or 6-DoF, sting balance was used. Because the internal measurements are mechanically coupled, the following 6x6 calibration matrix was used from Aerolab to decouple signals into forces and moments. This process removes mechanical coupling between channels so each output corresponds to a physical force or moment

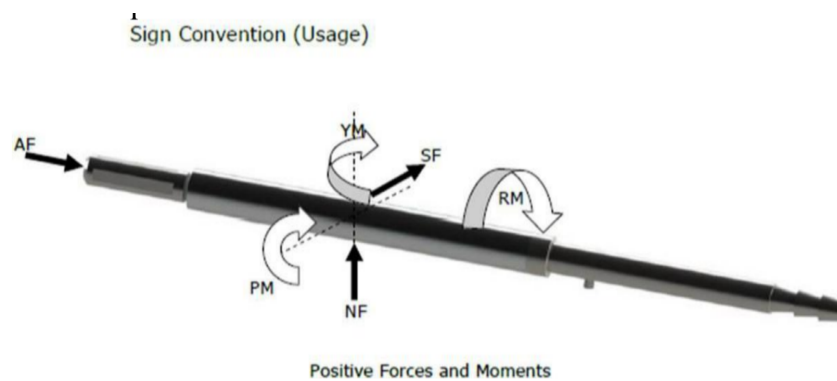


Figure 1.7: Model of Force Balance Used for Experimentation <sup>[7]</sup>

### Experiment 4: Boundary Layer

Whenever a fluid flows over a solid surface, viscous effects cause the fluid velocity at the wall to go to zero. This condition, known as the no-slip condition, comes from shear stresses between the fluid and the wall and is a consequence of fluid viscosity. As a result, the velocity near the surface is significantly lower than the freestream velocity, and a velocity gradient develops normal to the wall. The region over which the velocity transitions from zero at the wall to approximately the freestream value is called the boundary layer.

Within the boundary layer, viscous forces are dominant and the flow is rotational and often unsteady. Because of this, Bernoulli's equation cannot apply in the boundary layer region. Instead, the flow is defined by the full Navier-Stokes equations, which are difficult to solve in most situations. To make progress, boundary layer theory simplifies the Navier-Stokes equations by recognizing that velocity gradients normal to the wall are much larger than those parallel to it. This approach was first developed by Ludwig Prandtl and forms the foundation for modern aerodynamic analysis. <sup>[8]</sup>

As the flow continues downstream along the surface, its boundary layer thickness increases as the viscosity effects influence fluid farther away from the wall. The boundary layer thickness, referred to as  $\delta$ , is defined as the distance from the wall to the location where the local velocity reaches 99% of the freestream velocity. Initially, the boundary layer is laminar with a smooth flow and little exchange between fluid layers, but farther downstream, depending on the Reynolds number, the boundary layer undergoes a transition to turbulent flow. A turbulent boundary layer has a much steeper velocity gradient near the wall and more exchange between fluid layers, therefore having higher drag but more resistance to flow separation. <sup>[8]</sup>

Because the flow inside the boundary layer is rotational and viscous, it has a large impact in determining drag, heat transfer, and its separation behavior on real surfaces. A boundary layer's growth is therefore important for considering aerodynamic design as it influences aircraft wings, wind turbine blades, and ground transportation vehicles. For this experiment, boundary layer velocity profiles are measured using a rake placed normal to the surface in the wind tunnel. By measuring velocity as a function of distance from the wall at a fixed tunnel speed of 30 Hz, both upstream and downstream boundary layers can be visualized. Plotting rake position versus velocity allows the boundary layer's thickness and shape to be analyzed, which shows the physical effects of viscosity in the boundary layer.

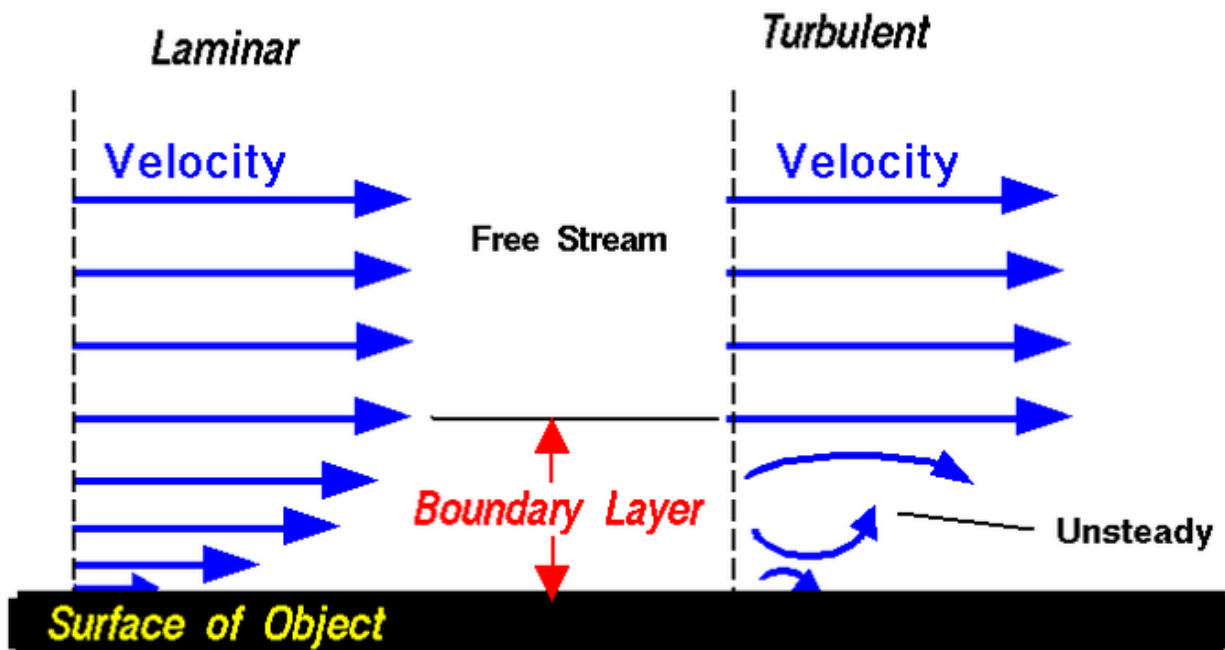


Figure 1.8: Boundary Layer Model Comparing Laminar vs. Turbulent Flow in a Wind Tunnel with Constant Free Stream Velocity

## Experiment 5: Airfoils

Airfoils are shapes that are designed to generate lift by directing the flow of air in a controlled manner.

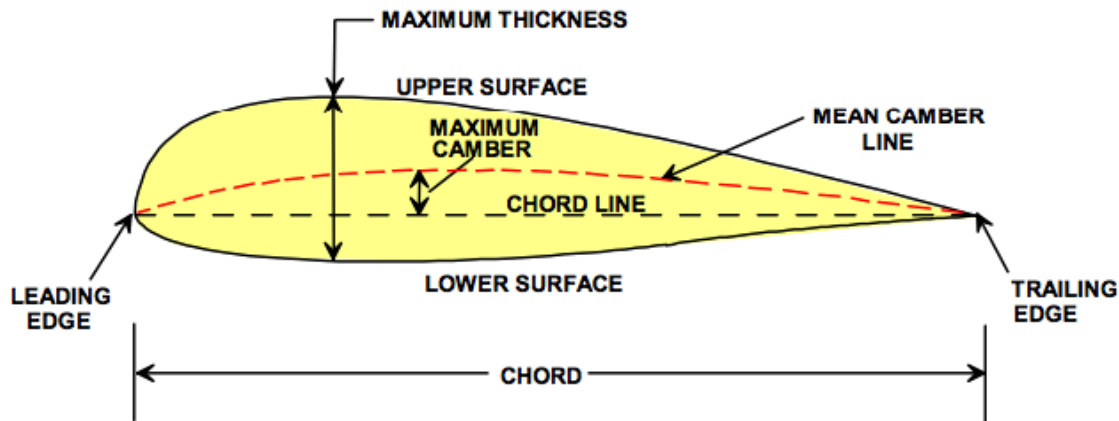


Figure 1.9: Airfoil Diagram Showing Common Terminology <sup>[9]</sup>

As fluid flows over an airfoil, viscous effects cause a boundary layer to form on the surface, the same way flow moves over any solid body. However, unlike bluff bodies, airfoils are shapes that are intended to maintain flow over varying conditions. The aerodynamic forces produced by an airfoil come from differences in the velocity and pressure between the upper and lower surfaces, which are determined by the Navier-Stokes equations and simplified using aerodynamic theory.

For a symmetric airfoil at zero angle of attack, the flow and boundary layers on the upper and lower surfaces are equal, resulting in no net lift. When the airfoil is rotated to a nonzero angle of attack, or when a cambered, or non-symmetric, airfoil is used, the velocity over one surface increases while decreasing on the opposite surface. This difference in velocity between both surfaces is defined by the momentum and pressure relationships described by the Navier-Stokes equations, showing that higher velocity corresponds to lower pressure and lower velocity corresponds to higher pressure. This pressure difference produces a net aerodynamic force known as lift.

Lift per unit area is known as the coefficient of lift and is expressed by the dimensionless parameter:

$$C_L = \frac{F_L}{\frac{1}{2}\rho U_\infty^2 A_p}$$

Where  $F_L$  is the lift force and  $A_p$  is the planform area of the airfoil. Thin airfoil predicts that, for a symmetric airfoil at small angles of attack, the coefficient of lift varies linearly with angle of attack, approximated by the equation below:

$$c_L = 2\pi\alpha$$

With  $\alpha$ , or angle of attack, expressed in radians. This relationship is known as the lift-curve slope, and is widely used in aerodynamic design.

Similarly, the force of drag,  $F_D$ , on an airfoil is described by the coefficient of drag:

$$c_D = \frac{F_D}{\frac{1}{2}\rho U_\infty^2 A_p}$$

Airfoils experience low pressure drag when the flow remains attached, unlike bluff bodies. However, as angle of attack increases, the pressure drag increases with it due to the increasing pressure differences and the increase in flow region separation. At high angles of attack, the boundary layer on the upper surface can no longer stay attached and begins to separate near the trailing edge. As separation progresses upstream, the airfoil begins to stall, which is seen when there is a sudden loss of lift and a sharp increase in drag.

In addition to lift and drag, airfoils experience a pitching moment,  $M$ , which is caused by the distribution of pressure around the chord. The moment is described using the coefficient of moment:

$$c_M = \frac{M}{\frac{1}{2}\rho U_\infty^2 A_p c}$$

Where  $c$  is the mean chord length. Thin airfoil theory says that the pitching moment about the point  $\frac{1}{4}c$ , or the quarter-chord point, remains near constant with the angle of attack and the moment is zero no matter the magnitude of lift, defining the mean aerodynamic center. This property is necessary for the stability of an aircraft and its control, as it influences how forces are applied to surfaces. For supersonic airfoils, the aerodynamic center is closer to half the chord location.<sup>[10]</sup>

Studying airfoil aerodynamics is necessary for its use in aircraft wings, propellers, and wind turbines. Understanding how lift, drag, and pitching moment vary with angle of attack and Reynolds number is important for designing efficient and stable aerodynamic systems. In this experiment, force and moment data are captured across a range of flow conditions in order to capture these trends and observe the effects of flow separation and stall. By comparing clean and

modified, also known as dirty, configurations, the experiment can show how the conditions of a surface have influence on the boundary layer's behavior and the airfoils overall aerodynamic performance.

### Experiment 6: 2-D vs. 3-D Objects

Flow over an object in a wind tunnel may be approximated as either two-dimensional or three-dimensional depending on the object's span relative to the test section. When a body spans the full width of the tunnel, it is treated as having infinite span and the flow is assumed to be two-dimensional. Objects with finite length produce three-dimensional flow due to the side edges, where additional boundary layer separation occurs.

The amount of flow separation is the main differentiating factor between both dimensions. In two-dimension flow over a bluff body such as a cylinder, separation happens mainly on the top and bottom surfaces, creating wake downstream. In three-dimensional flow, separation also occurs along the sides, increasing the wake size and pressure difference between the front and end of the object. This leads to higher pressure drag on the object.

Three-dimension objects also experience additional energy loss due to trailing vortices that form at the ends of the object, which are not seen in two-dimensional flow. These vortices increase momentum loss in the wake and further increase drag. These effects are important for real world applications such as bridge cables where finite length and end effects strongly influence aerodynamic performance. In this experiment, cylinders of varying lengths are tested at the same flow conditions to look at the effects of three-dimensional increases in drag compared to a two-dimensional case.

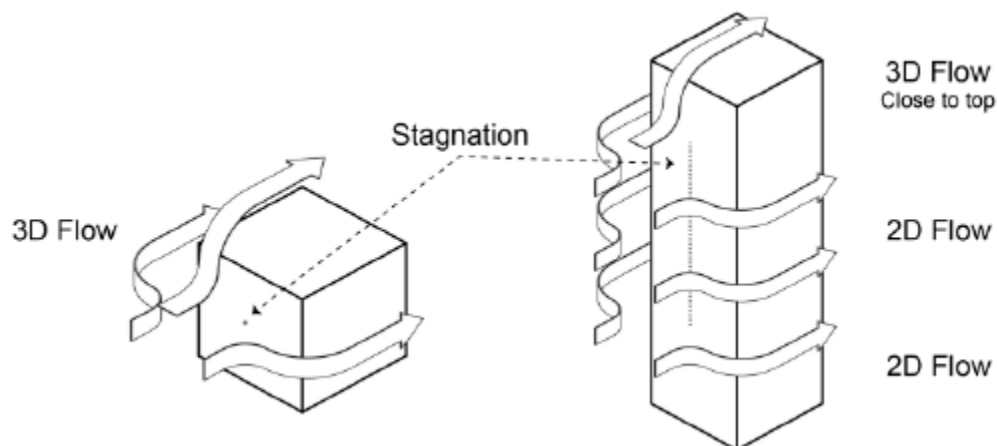


Figure 1.10: Stagnation Differences in 2D vs. 3D Flow <sup>[11]</sup>

## Procedure

### Equipment

- Closed loop wind tunnel
- Pitot-static tube
- Aerolab 6 DoF (Degrees of Freedom) Sting Balance
- Clark Y airfoil
- Daytronic System 10 DataPac
- Pressure Systems 9010 Optomux
- Pitot tube rake
- Accelerometer
- 3x 2" O.D. Cylinder
- 5x 4.8cm O.D. Cylinder
  - Lengths: 10.5", 18", 20", 22", and 24"
- Thermometer
- LabVIEW software
- Ruler
- Calipers

### Experiment 1: Wind Tunnel Calibration

Experiment 1 established a relationship between the wind tunnel fan frequency and freestream velocity. A Pitot-static tube was mounted at the center of the test section and aligned with the freestream to measure pressure differences caused by airflow. With the wind tunnel initially turned off, pressure readings and temperature were recorded to create a baseline reference for the pressure transducer. The tunnel speed was then increased incrementally from 5 Hz up to 40 Hz in increments of five. At each fan frequency, multiple pressure readings were recorded along with the air temperature. These pressure measurements can then be converted into velocity using Bernoulli's equation, allowing fan frequency to be correlated with flow speed.

### Experiment 2: Vortex shedding

Experiment 2 measured vortex shedding frequency from a circular cylinder using vibrations. The cylinder diameter and length were measured first to later find Reynolds number. The cylinder was then mounted vertically in the test section, with an accelerometer attached to detect oscillations from the vortex shedding. The wind tunnel was run at 30 Hz and data was connected using LabVIEW at a high sampling frequency of 6400 Hz and sampled 16384 times total to capture unsteady forces. This procedure was then repeated at 45 Hz. The accelerometer's data was later analyzed to find the most dominant vortex shedding frequency.

### Experiment 3: Wake

Experiment 3 looks at velocity and pressure changes downstream of a bluff body. The spacing between pressure taps on the rake is first measured to determine spatial resolution. The rake was first placed in the wind tunnel without a cylinder present, and pressure data was recorded at two tunnel speeds to establish freestream conditions. The cylinder was then mounted

upstream of the rake at a distance of eight diameters to make sure the wake was fully developed when reaching the measured location. Pressure data was once again recorded multiple times at each rake position and averaged to reduce noise. These measurements were used to determine the wake velocity profile and loss in momentum.

#### Experiment 4: Boundary Layer

Experiment 4 measured the velocity profile within the boundary layer over a flat surface. The wind tunnel was operated at 30 Hz. A boundary layer rake was placed at the center of the test section with all tubes parallel to the flow. Each tube was connected to a pressure scanner, and pressure data was acquired using LabVIEW. The air temperature was recorded manually using a thermometer to find the air density. The rake was then moved to a second location, and the measurements were repeated. These velocity profiles were used to determine boundary layer thickness upstream and downstream.

#### Experiment 5: Airfoil

Experiment 5 measured lift, drag, and pitching moment as a function of angle of attack. A Clark Y airfoil was mounted on a six-degree-of-freedom sting balance in the wind tunnel. The tunnel was operated at both 30 Hz and 45 Hz. Starting at a negative angle of attack, force and moment data were recorded at each angle using LabVIEW, and the temperature was noted after every run. The angle of attack was increased incrementally until stall was seen, indicated by flow separation on the tufts attached to the airfoil. After completing the clean data collection, flaps and slats were used and the measurements were repeated. Airfoil dimensions were measured at the end for later calculation of aerodynamic coefficients.

#### Experiment 6: 2-D vs. 3-D Objects

Experiment 6 compared drag forces on cylinders of varying lengths to study the difference in flow between two and three dimensions. The length and diameter of each cylinder were measured before testing. A baseline measurement was taken with no cylinder to account for any tunnel drag. Each cylinder was then mounted perpendicular to the flow and tested at a tunnel speed of 30 Hz. Drag forces and temperature were recorded using the force balance system. The process was repeated for all cylinders. Comparing the resulting drag coefficients on different cylinder lengths can allow for observation of the influence of three-dimensional end effects.

## Results

### Experiment 1: Wind Tunnel Calibration

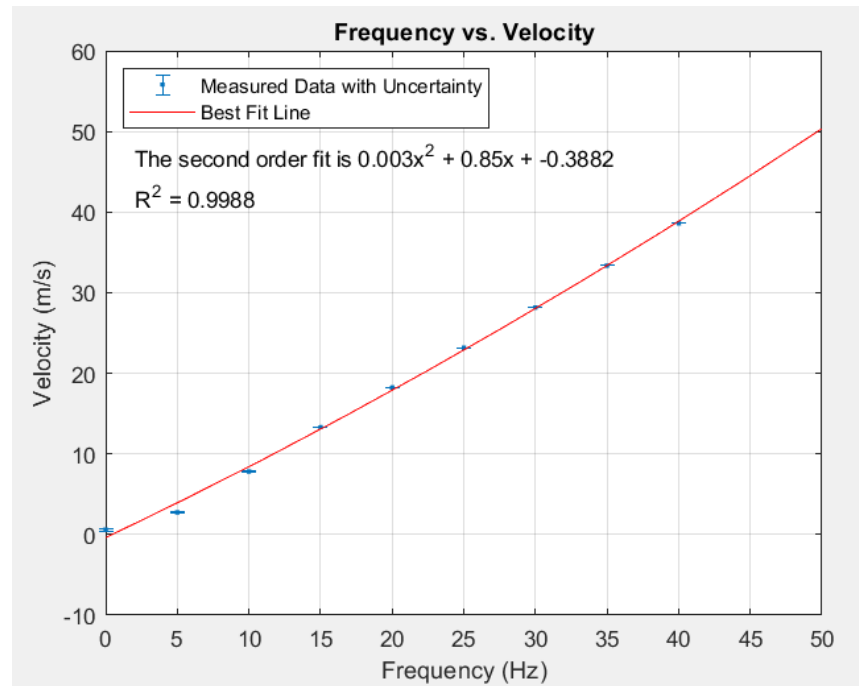


Figure 3.1.1: Wind Tunnel Frequency vs. Wind Tunnel Velocity With Uncertainties

Figure 3.1.1 shows the relationship between wind tunnel fan frequency and freestream velocity measured by a pitot-static tube. This plot has the measured velocities with their associated uncertainty bars, alongside with a second-order polynomial fit to the data and the coefficient of determination. As hypothesized, velocity increases with fan frequency, indicating that higher fan speeds generate larger pressure differences and therefore a higher flow velocity.

From fluid mechanics, the freestream velocity relative to pressure difference in Bernoulli's equation:

$$U = \sqrt{\frac{2\Delta P}{\rho}}$$

Where  $\Delta P$  is the dynamic pressure measured by the pitot tube and  $\rho$  is the air density. Since pressure rise by the fan increases with a faster fan speed, we can derive an equation of near linear relationship between frequency and velocity in the wind tunnel. The small quadratic term likely shows slight variation in the tunnel. The line of best fit curve is shown with the equation:

$$U = 0.003x^2 + 0.85x - 0.3882$$

The coefficient of determination shows the goodness of fit as  $R^2$  of 0.9988. This means that the calibration equation explains nearly all variation in the measured velocity data. The data

points lie very close to the fit line, showing that the measurements are consistent and calibration is reliable.

The uncertainty bars on the plot represent the uncertainty in velocity likely due to pressure sensor resolution. The uncertainty is greater at lower frequencies because there are such small pressure differences, but as the frequency increases, the relative uncertainty in velocity becomes smaller, which is why the error decreases.

Table 3.1.1: Wind Tunnel Testing with Incremental Frequency Increase of 5 Hz

| Velocity ( <i>m/s</i> ) $\pm$ uncertainty | Frequency ( <i>Hz</i> ) $\pm$ accelerometer uncertainty |
|-------------------------------------------|---------------------------------------------------------|
| $0.61 \pm 1.8 \cdot 10^{-1}$              | $0 \pm 0.5$                                             |
| $2.75 \pm 4.1 \cdot 10^{-2}$              | $5 \pm 0.5$                                             |
| $7.82 \pm 1.4 \cdot 10^{-2}$              | $10 \pm 0.5$                                            |
| $13.32 \pm 8.4 \cdot 10^{-3}$             | $15 \pm 0.5$                                            |
| $18.25 \pm 6.1 \cdot 10^{-3}$             | $20 \pm 0.5$                                            |
| $23.16 \pm 4.8 \cdot 10^{-3}$             | $25 \pm 0.5$                                            |
| $28.17 \pm 4.0 \cdot 10^{-3}$             | $30 \pm 0.5$                                            |
| $33.35 \pm 3.4 \cdot 10^{-3}$             | $35 \pm 0.5$                                            |
| $38.61 \pm 2.9 \cdot 10^{-3}$             | $40 \pm 0.5$                                            |

Table 3.1.2: Freestream Velocity Predictions from Wind Tunnel Calibration at 40 Hz and 45 Hz.

| Frequency ( <i>Hz</i> ) | 40    | 45    |
|-------------------------|-------|-------|
| Velocity ( <i>m/s</i> ) | 38.85 | 44.52 |

Using the wind tunnel calibration curve, the freestream of velocity at 40 Hz is 38.85 m/s. Applying the same equation to 45 Hz gives a prediction of 44.52 m/s. This velocity at 45 Hz is derived through extrapolation of the calibration curve beyond the highest measured frequency. Because of the previous linear trend, the velocity at 45 Hz is reasonable, however it should be taken into consideration that there is slightly less confidence in any value above the measured 40 Hz.

## Experiment 2: Vortex Shedding

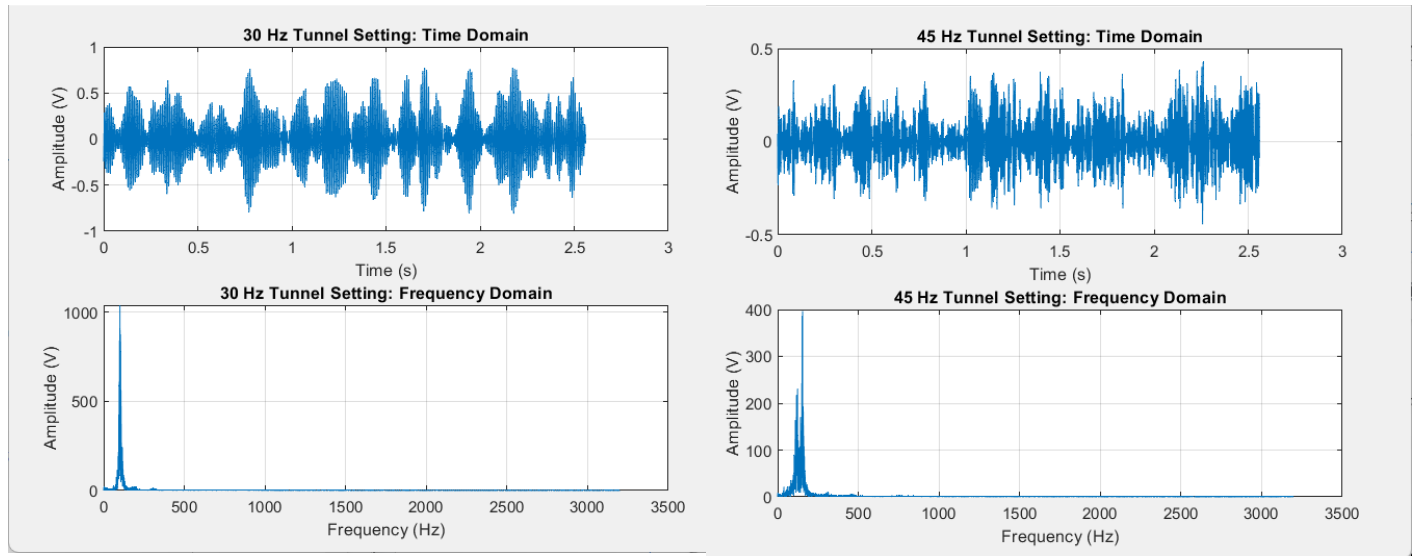


Figure 3.2.1 (left) and 3.2.2 (right): Time and Frequency Domain Accelerometer Results of a Circular Cylinder at 30 Hz and 45 Hz Wind Tunnel Speeds

The amplitude versus time plots of the 30 Hz and 45 Hz tunnel runs show oscillation of the accelerometer. This tells us that the cylinder vibrated in a repeated pattern rather than randomly. The amplitude versus frequency plots show sharp peaks at specific frequencies of roughly 100 Hz and 151 Hz for the 30 Hz and 45 Hz runs, respectively, representing the primary vortex shedding frequency.

According to fluid mechanics theory, when a flow passes a bluff body like a circular cylinder, vortices are shed alternating from either side, forming Von Kármán vortex streets. A fast Fourier transform, or FFT, is used to get the peaks seen, which represent these oscillations on the cylinder, seen proportional to the freestream velocity.<sup>[13]</sup>

Table 3.2.1: Vortex Shedding Data and Flow Results for a Circular Cylinder at 30 Hz and 45 Hz.

| Tunnel Frequency (Hz)   | 30 Hz                                       | 45 Hz                                       |
|-------------------------|---------------------------------------------|---------------------------------------------|
| Velocity (m/s)          | $28.04 \pm 3.99 \cdot 10^{-3}$              | $44.52 \pm 2.52 \cdot 10^{-3}$              |
| Shedding Frequency (Hz) | $100.39 \pm 1.95 \cdot 10^{-1}$             | $151.17 \pm 1.95 \cdot 10^{-1}$             |
| Strouhal Number         | $1.74 \cdot 10^{-1} \pm 3.39 \cdot 10^{-4}$ | $1.65 \cdot 10^{-1} \pm 2.13 \cdot 10^{-4}$ |
| Reynolds Number         | $9.07 \cdot 10^4 \pm 12.91$                 | $1.43 \cdot 10^5 \pm 8.13$                  |

|                          |                                 |                                 |
|--------------------------|---------------------------------|---------------------------------|
| Empirical Frequency (Hz) | $106.48 \pm 1.51 \cdot 10^{-2}$ | $106.48 \pm 9.58 \cdot 10^{-3}$ |
|--------------------------|---------------------------------|---------------------------------|

From the FFT results, we see that the 30 Hz and 45 Hz run have a shedding frequency of about  $100.39 \pm 1.95 \cdot 10^{-1}$  and  $151.17 \pm 1.95 \cdot 10^{-1}$ , respectively. These values are consistent with the expected behavior that frequency increases with velocity in vortex shedding.

We are able to use the same accelerometer as would be used in a lab measuring vibrations as the accelerometer measures motion and not the source of said motion. In a vibrations lab, it measures mechanical oscillations while in this lab, it detects fluid oscillations. In both cases, there is a periodic signal and an FFT is used to find the primary frequency. Because vortex shedding is periodic on the cylinder, the accelerometer can capture shedding frequency.

As previously defined, the Strouhal number relates vortex shedding frequency, flow speed, and body size in the equation:

$$St = \frac{f_s D}{U}$$

For the flows measured, the Strouhal number is about constant, only fluctuating by 0.01 over a wide range of Reynolds numbers, meaning that shedding frequency scales with velocity. The Reynolds number also increases with velocity, showing that bluff body shedding is expected. The results from Table 3.2.1 are expected, as the shedding frequency increases with the velocity while the Strouhal number stays relatively constant.<sup>[13]</sup>

### Experiment 3: Cylinder Wake

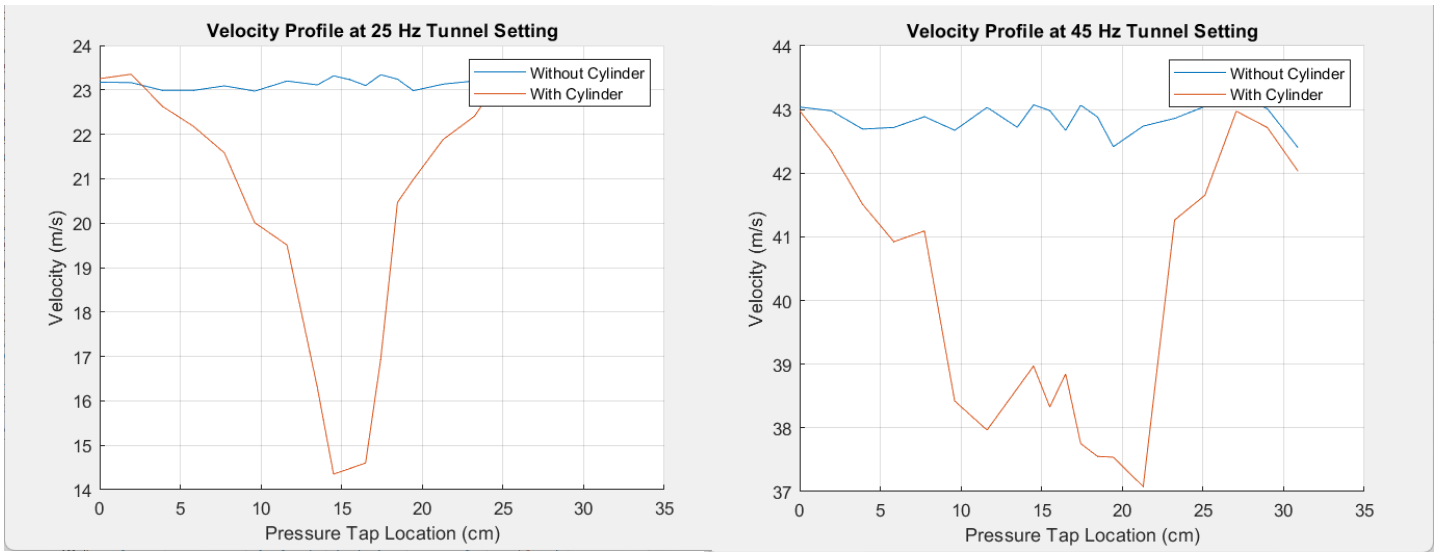


Figure 3.3.1 (left) and 3.3.2 (right): Velocity Profiles vs. Pressure Tap Locations at 25 Hz and 45 Hz, With and Without a Cylinder Present

Dry air properties at room temperature were used, with dynamic viscosity of  $1.79 \cdot 10^{-5} \frac{kg}{m \cdot s}$ . Using these properties allows the flow in the tunnel to be compared to published cylinder wake data.

Table 3.3.1: Wake drag, coefficient of drag, and Reynolds Number for 25 Hz and 45 Hz tunnel speeds.

| Frequency of tunnel (Hz)                 | 25                                    | 45                                          |
|------------------------------------------|---------------------------------------|---------------------------------------------|
| Drag from cylinder $\pm$ uncertainty (N) | $11.17 \pm 1.23 \cdot 10^{-1}$        | $25.28 \pm 7.40 \cdot 10^{-2}$              |
| Coefficient of Drag $\pm$ uncertainty    | $1.21 \pm 2.99 \cdot 10^{-2}$         | $7.25 \cdot 10^{-1} \pm 4.75 \cdot 10^{-1}$ |
| Reynolds Number $\pm$ uncertainty        | $7.40 \cdot 10^4 \pm 8.18 \cdot 10^2$ | $1.43 \cdot 10^5 \pm 4.20 \cdot 10^2$       |

These values were calculated with the previously derived equations of:

$$F_D = \int_0^W \rho u_2 (u_1 - u_2) dt \quad \& \quad C_D = \frac{F_D}{\frac{1}{2} \rho U_\infty^2 A_p}$$

Reynolds number was calculated from published density and viscosity with:

$$Re = \frac{\rho V D}{\mu}$$

It is important to mention that tap #10 was excluded from these results when measuring at 25 Hz and tap #8 was removed when measured at 45 Hz. These taps both were unable to accurately measure the velocity and instead were replaced with an interpolation of where the data should be.

The velocity versus pressure tap location plots show a difference between freestream flow and flow with the cylinder. Without the cylinder, the velocity is relatively constant through the test, while when the cylinder is present, a wake forms which is seen by the reduced velocity caused by the cylinders surface. The drag coefficients calculated from the wake are consistent with published data for cylinders at similar Reynolds numbers. A smooth cylinder's coefficient of drag typically falls around 1,<sup>[12]</sup> which proves this experiments margin of error to be approximately  $\pm 0.3$ . Error could be caused due to inaccuracies in the rake taps, software error, or simple measurement uncertainty accumulation.

## Experiment 4: Boundary Layer

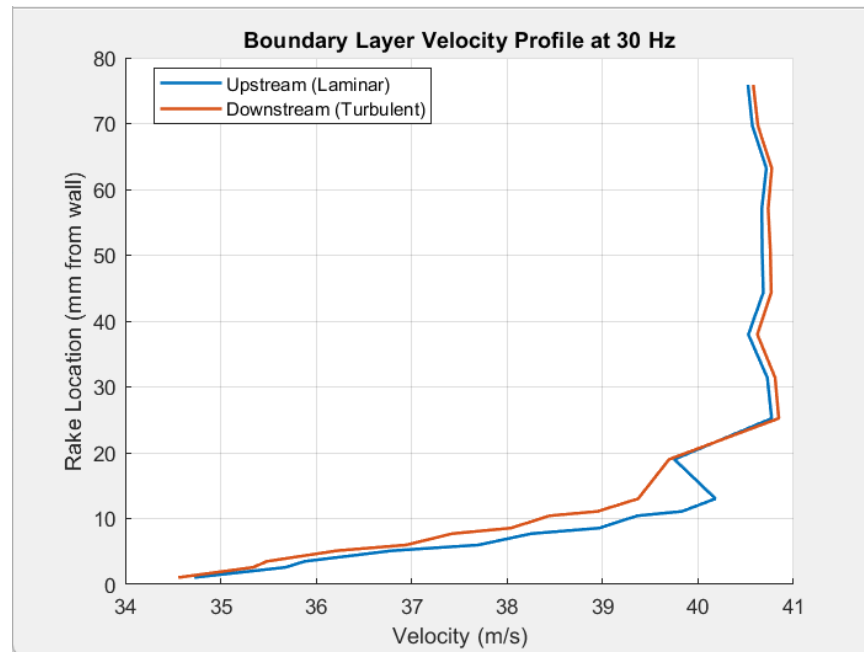
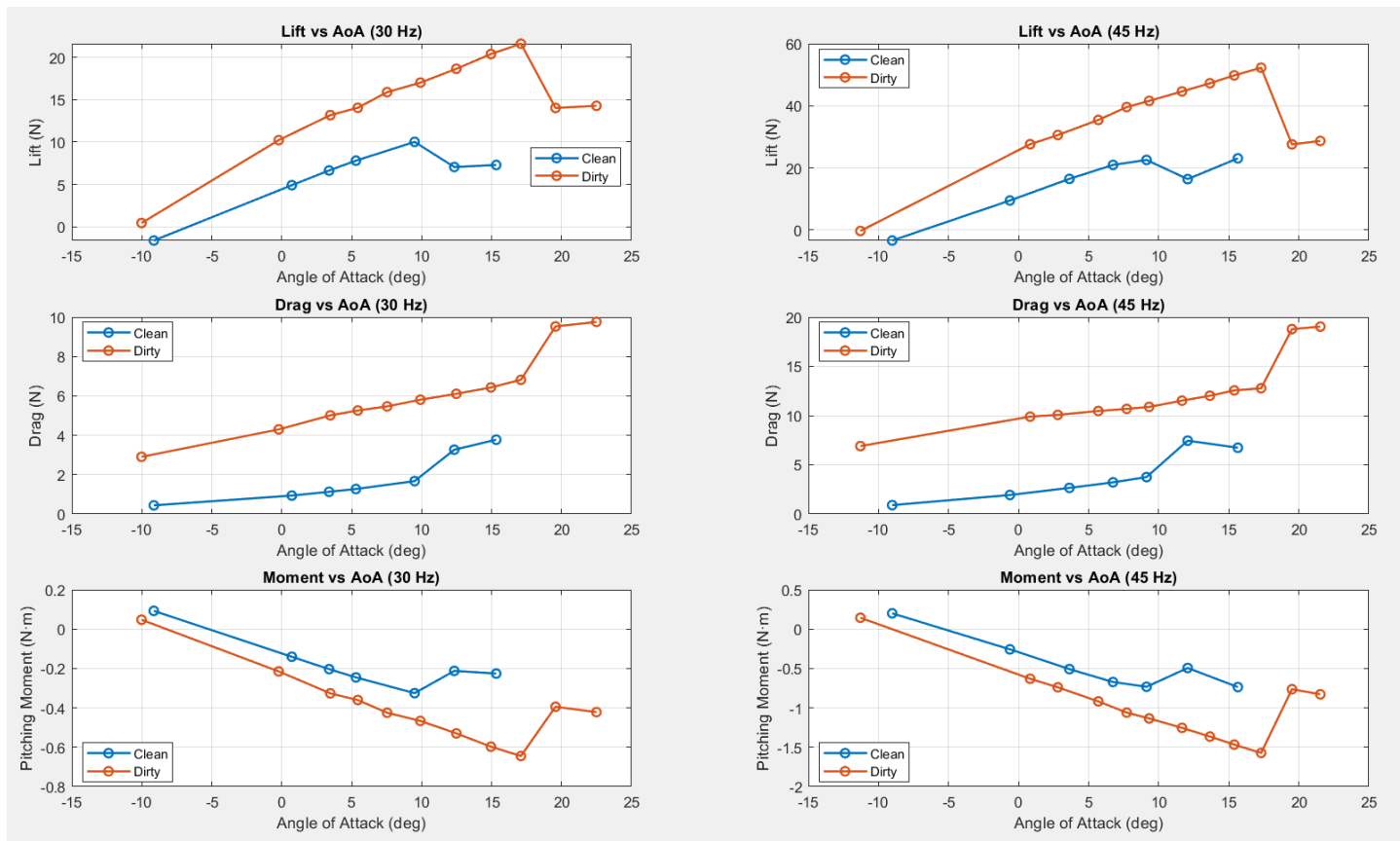


Figure 3.4.1: Boundary Layer Velocity Profile at 30 Hz

The boundary layer velocity profiles show how the airspeed changes from the wall out to the freestream. Near the wall, the velocity is low because of the no-slip condition, which means the air molecules at the surface have zero velocity relative to the wall. As you move away from the wall, the velocity increases until it approaches the freestream value. This behavior is exactly what the boundary layer theory predicts, that viscous effects near the surface slow the flow while farther away the flow is dominated by inertia rather than viscosity.

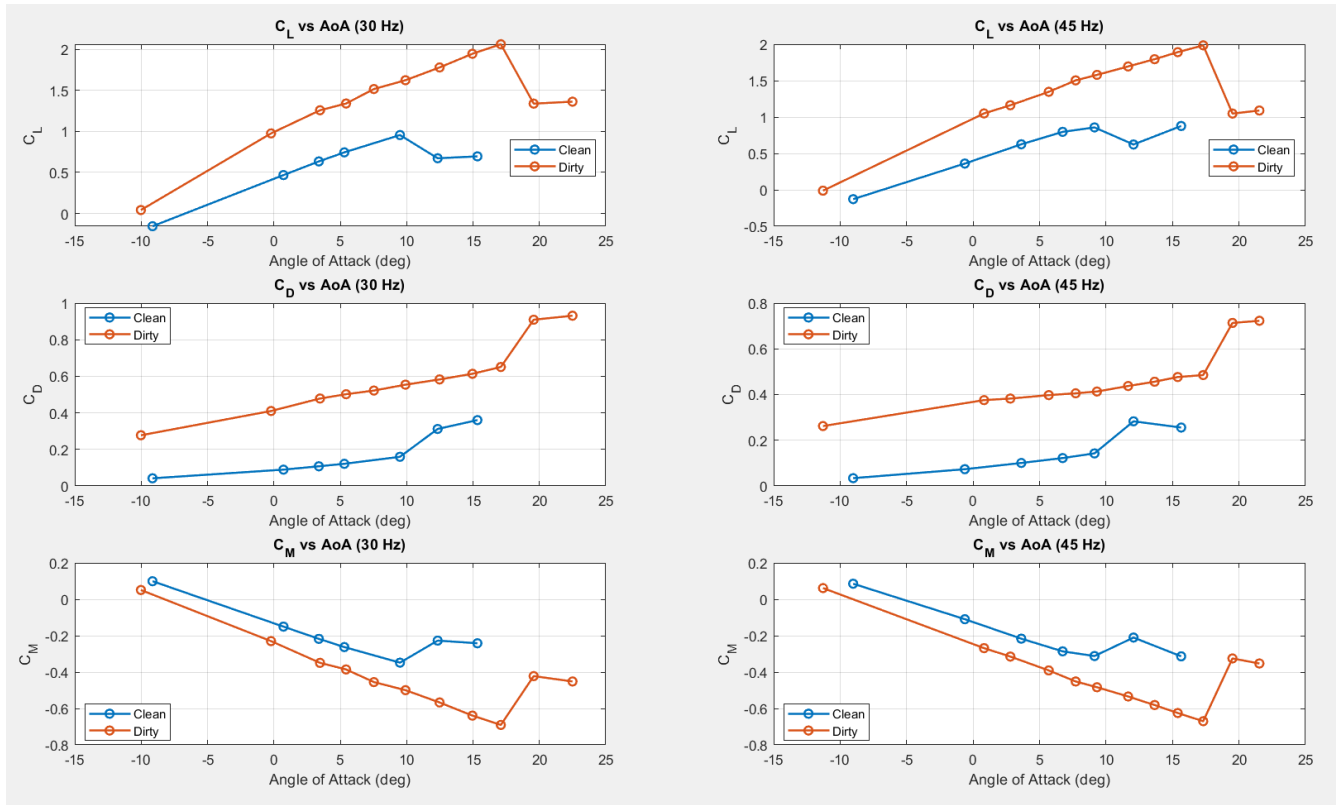
Comparing the upstream and downstream profiles at 30 Hz, the downstream boundary layer is slightly fuller with higher velocity closer to the wall. This is because the boundary layer grows in thickness as the flow moves downstream, and turbulent boundary layers mix momentum better than laminar ones. As the layer grows, the velocity profile adjusts so the mass is conserved across the cross section. The freestream velocity does not change significantly with streamwise distance, which ties to the idea that most of the velocity change in a boundary layer happens close to the surface rather than in a free stream. The measured profile flow is the expected boundary layer development: low velocity at the wall that transitions into freestream velocity and a growing boundary layer thickness downstream.

## Experiment 5: Airfoil



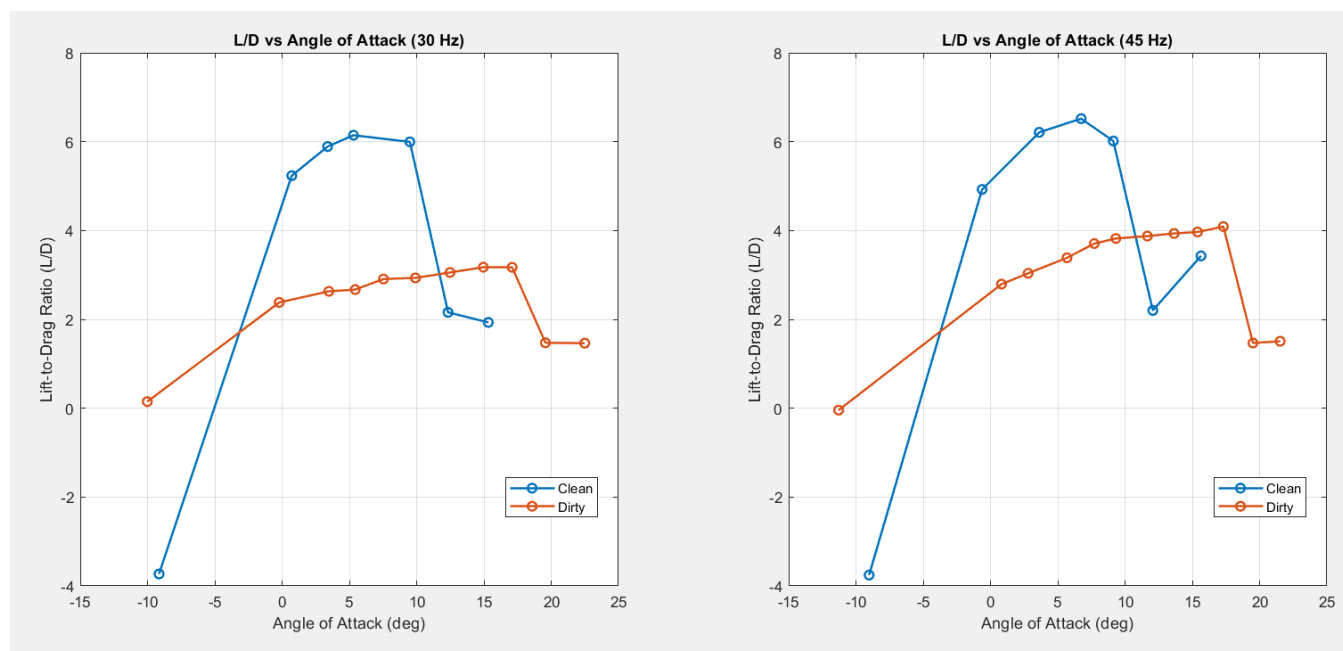
Figures 3.5.1 - 3.5.6: Angle of Attack vs. Lift, Drag, and Pitching Moment (30 Hz and 45 Hz, Clean and Dirty)

These figures show how the measured aerodynamic forces and pitching moment vary based on the angle of attack for both tunnel speeds and surface conditions. Lift increases linearly with angle of attack at small angles, while drag increases as the angle of attack increases. The pitching moment decreases linearly with angle of attack and shows changes in pressure over the airfoil. The dirty configuration produces higher lift and drag compared to the clean case due to increased surface roughness.



Figures 3.5.7 - 3.5.12: Angle of Attack vs. Coefficient of Lift, Drag, and Pitching Moment (30 Hz and 45 Hz, Clean and Dirty)

These plots show the nondimensional coefficients which allows comparison between them and Reynolds numbers. The coefficient of lift and drag show a positive linear region at low angles of attack and followed by deviation near stall while the coefficient of moment has a negative linear region.



Figures 3.5.13 and 3.5.14: Angle of Attack vs. Lift-to-Drag ratio (30 Hz and 45 Hz)

These plots show how variation in angle of attack impacts aerodynamic efficiency, as the lift-to-drag ratio increases to the maximum at moderate angles of attack before decreasing as drag rises rapidly near stall. The clean configuration has higher peak efficiency than dirty.

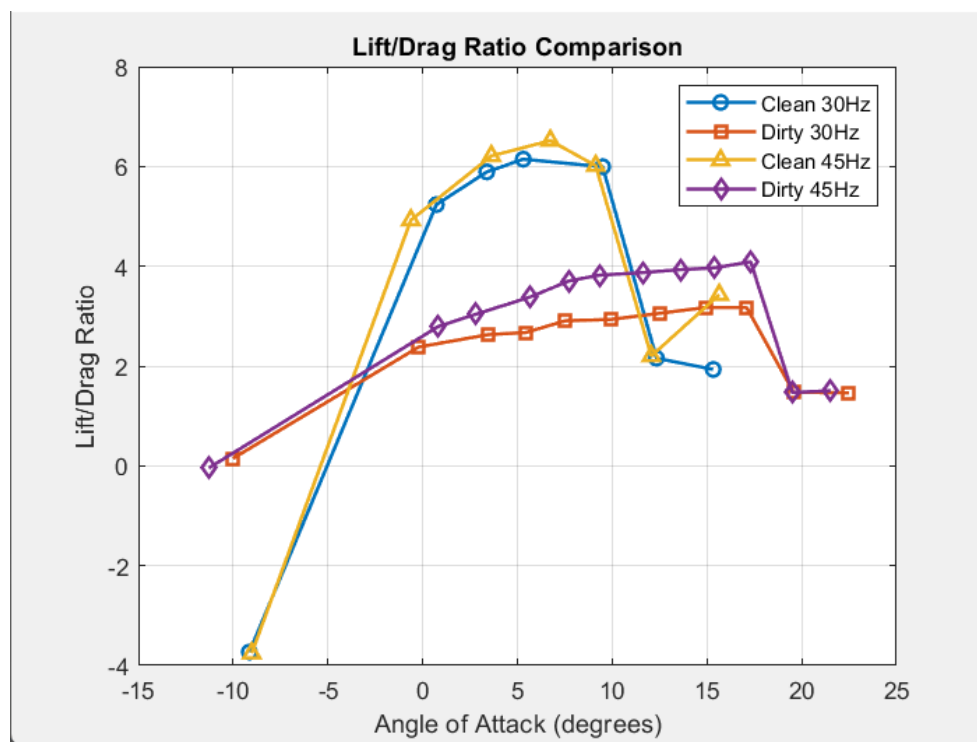


Figure 3.5.15: Lift-to-Drag Ratio Comparison for All Configurations

This figure compares lift-to-drag ratios for all four cases of 30 Hz and 45 Hz between clean and dirty all on one plot. The clean 45 Hz flow has the highest overall efficiency while dirty configurations show reduced performance all around.

Table 3.5.1: Airfoil Performance at 30 Hz and 45 Hz Fan Frequencies

| Fan Frequency ( <i>Hz</i> ) | 30                            | 45                            |
|-----------------------------|-------------------------------|-------------------------------|
| Reynolds Number             | 166,457                       | 263,475                       |
| Peak Efficiency (Clean)     | 6.15 at $\alpha = 5.3^\circ$  | 6.52 at $\alpha = 6.7^\circ$  |
| Peak Efficiency (Dirty)     | 3.17 at $\alpha = 15.0^\circ$ | 4.09 at $\alpha = 17.3^\circ$ |
| Lift-Curve Slope (Clean)    | 0.0625 per degree             | 0.0623 per degree             |
| Lift-Curve Slope (Dirty)    | 0.0765 per degree             | 0.0568 per degree             |

The lift curves for both tunnel speeds show an approximately linear relationship between the lift coefficient and angle of attack when at low angles. The theoretical lift-curve slope for a Clark-Y airfoil is  $2\pi$  per radian, or about 0.11 per degree, and the measured slopes for the clean configurations are not particularly close due to finite wing effects and tunnel wall interference. They are noticeably lower than the ideal theoretical value. This could be because of software error, tunnel wall interference, or airflow problems stemming from the wind tunnel itself.

Tuft visualization indicates that the flow separation began at an angle of attack of approximately  $12^\circ$ . This is consistent with the force data as, in a clean configuration, lift begins to level off at about  $10^\circ$  while in a dirty configuration, that is closer to  $17^\circ$ . The higher separation angle is expected because slats give energy to the boundary layer which allows flow to remain attached longer and delay stall.

Tufts separate from the surface of an airfoil due to stall. As the angle of attack increases, the pressure gradient on the upper surface becomes stronger and eventually the boundary layer does not have enough momentum to remain attached, and the flow separates. When this happens, lift decreases rapidly while drag increases significantly. This flow separation causes the tufts to lift off the surface and fluctuate, which visually shows stalled flow. Stall is a critical condition for airfoils because it leads to a loss in lift and reduced control, and explains the change seen in the lift-to-drag ratio.

The Reynolds number does affect the lift, drag, and pitching moment of the airfoil as higher Reynolds numbers correspond to higher freestream velocities. This increases the inertial forces relative to the viscous forces in the boundary layer. As Reynolds number increases, the boundary layer can better resist separation which results in higher lift, lower drag, and changes in the pitching moment. Although Reynolds number does not explicitly appear in force equations, it influences the flow and determines the forces used to compute lift, drag, and moment.

Changing the angle of attack can slightly affect the freestream velocity in the wind tunnel because as the airfoil rotates, it changes the effect of flow area within the test section. Mass flow rate has to remain constant so there are changes in velocity. These changes are very small but become more noticeable near stall as the boundary layer thickens and flow separation increases.

## Experiment 6: 2D-3D Cylinder

Table 3.6.1: Effect of Cylinder Length on Reynolds Number, Drag, and Drag Coefficient

| Cylinder                              | <i>Short (3)</i>               | <i>Medium (2)</i>              | <i>Long (1)</i>                |
|---------------------------------------|--------------------------------|--------------------------------|--------------------------------|
| Diameter ( <i>mm</i> )                | 48.03                          | 48.30                          | 48.25                          |
| Length ( <i>cm</i> )                  | 26.7                           | 50.7                           | 60.4                           |
| Reynolds Number $\pm$ uncertainty     | $2.22 \cdot 10^4 \pm 36.57$    | $2.21 \cdot 10^4 \pm 36.78$    | $2.20 \cdot 10^4 \pm 36.74$    |
| Drag ( <i>N</i> ) $\pm$ uncertainty   | $6.79 \pm 1.36 \cdot 10^{-1}$  | $13.41 \pm 2.68 \cdot 10^{-1}$ | $16.37 \pm 3.27 \cdot 10^{-1}$ |
| Coefficient of Drag $\pm$ uncertainty | $18.90 \pm 3.85 \cdot 10^{-1}$ | $19.54 \pm 3.97 \cdot 10^{-1}$ | $20.03 \pm 4.07 \cdot 10^{-1}$ |

Table 3.6.1 shows how cylinders with near identical diameters but differing lengths will show the influence of three-dimensional end effects to be seen. The long cylinder had the highest coefficient of drag. This likely occurred because a longer cylinder allows for a greater pressure force in its wake, which will increase the drag coefficient.

In two-dimensional flow, the body is assumed to extend infinitely in span, so flow does not change along the length of the cylinder. This leads to uniform wake and a drag coefficient that depends mostly on Reynolds number. In three-dimensional flow, finite cylinder length introduces end effects, where flow near the ends moves around cylinder tips which generates streamwise vortices that change the wake. Although end effects become more significant for shorter cylinders, the longer cylinders produce larger wake and pressure deficit which will induce more drag.

Cylinder 1 has a substantially higher coefficient of drag than was seen in experiment three. Differences could be caused by measurement uncertainty or inconsistent flow from the tunnel. When compared with published data for a smooth circular cylinder with a similar Reynolds number range, the drag coefficient is often found to be around 1.0 - 1.3. Measured values here are noticeably higher, possibly due to end effects or error in LabVIEW software.

## Conclusions

This lab showed how fluid mechanics concepts directly appear in experimental wind tunnel measurements. The wind tunnel calibration established a reliable relationship between fan frequency and freestream velocity, which allowed for velocities at higher frequencies to be estimated through extrapolation. Vortex shedding measurements showed that shedding frequency increased with velocity while the Strouhal number remained nearly constant, which is consistent with theory for circular cylinders. Wake measurements confirmed that drag is caused by a loss of momentum in the wake, and drag coefficients fell within the expected range for bluff bodies of similar Reynolds numbers.

Boundary layer measurements showed the expected velocity behavior near a wall, with low velocities at the surface due to the no-slip condition and increasing velocity towards the freestream. The downstream boundary layer was thicker than the upstream profile, which matched predictions. Airfoil testing showed a linear lift region at low angles of attack, followed by stall at approximately 12 degrees as indicated by tuft separation. Higher Reynolds numbers improved aerodynamic performance by delaying separation and increasing the lift-to-drag ratio. The comparison between two and three dimensional cylinders showed that finite length effects increase drag due to additional flow separation and end vortices.

Overall, these results had varying accuracy but reinforced the role of viscosity, Reynolds number, and flow separation in real aerodynamic flow. To improve the lab, simply allocating more funding into the wind tunnel's software and hardware, such as upgrading the tunnel's data acquisition computer and the broken pressure rake in the hardware. Another way to improve it would be to explain the purpose of the data that is being collected in the lab instead of simply measuring data points from a computer.

## Appendix

[1]: Erdman, Dr. *Lab 5a: Fluid Dynamics Week 1*. Blackboard Handout provided for MAE 315. Syracuse University, 2025.

- *This handout was used for the majority of theory for Experiments 1-3.*

[2]: Roth, Samuel. *Why Not Blender? Comparative Study of Blender and ANSYS for Educational Airflow Modeling*. University College of Dublin, School of Engineering | Supervised by Dr. Kevin Nolan. 2025.

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[5]: Summary Planet. "Bluff Body Viscous Flow Characteristics." <https://www.summaryplanet.com/engineering/Bluff-Body-Viscous-Flow-Characteristics.html>.

[6]: Erdman, Dr. *Lab 5 Introduction Slides: Fluid Dynamics*. MAE 315: Mechanical And Aerospace Engineering Lab. Syracuse University.

- *This handout was used for the majority of theory for Experiments 4-6.*

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[https://www.researchgate.net/figure/Airfoil-shape-Chord-line-The-straight-line-connecting-the-leading-and-trailing-edges\\_fig2\\_362054845](https://www.researchgate.net/figure/Airfoil-shape-Chord-line-The-straight-line-connecting-the-leading-and-trailing-edges_fig2_362054845). Accessed 16 December 2025.

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[https://www.researchgate.net/figure/Flow-pattern-over-3D-left-and-2D-bodies-right\\_fig2\\_267776916](https://www.researchgate.net/figure/Flow-pattern-over-3D-left-and-2D-bodies-right_fig2_267776916).

[12]: Mendez, Miguel Alfonso. “RUNNING FINEOPEN43 SIMULATIONS AT VKI: A TUTORIAL AND A COLLECTION OF SCRIPTS.” ResearchGate,

[https://www.researchgate.net/publication/316588041\\_Running\\_FineOpen43\\_Simulations\\_at\\_VKI\\_A\\_tutorial\\_and\\_a\\_collection\\_of\\_scripts](https://www.researchgate.net/publication/316588041_Running_FineOpen43_Simulations_at_VKI_A_tutorial_and_a_collection_of_scripts).

[13]: Wikipedia. *Strouhal number - Wikipedia*,

[https://en.wikipedia.org/wiki/Strouhal\\_number?utm\\_source=chatgpt.com](https://en.wikipedia.org/wiki/Strouhal_number?utm_source=chatgpt.com). Accessed 16 December 2025.

*ChatGPT-5 was used in the writing and correction of grammar in this report. The prompt that was used is as follows:*

“Please proofread the selection for spelling and grammar errors. Identify any mistakes and provide a corrected version of the text. Maintain the meaning and factual accuracy and output the list of proposed corrections first, followed by the final, corrected version of the text.”